WEEK 5: REPRODUCIBILITY AUDIT REPORT

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TASK 1: THESIS MODEL COMPARISON

1. Selected Models

Random Forest was selected as the candidate thesis model because the doctoral research focuses on predicting student dropout using multiple academic, administrative, contextual, and temporal variables. This algorithm can capture nonlinear relationships and interactions among predictors, requires relatively few distributional assumptions, and provides feature importance estimates that can support interpretation. Although it does not represent the complete explainable, multimodal, and temporal framework proposed for the thesis, it provides an appropriate initial approach for evaluating its predictive component.

Logistic Regression was selected as the main baseline because it is a simple, transparent, and widely used classification method. A Dummy Classifier was also included to establish the minimum performance expected from an approach that does not learn meaningful predictive patterns. All three models were evaluated using the same data partitions and performance procedure to ensure a fair comparison.

2. Results

Model	Mean accuracy
Dummy Classifier	0.627 ± 0.004
Logistic Regression	0.978 ± 0.014
Random Forest	0.961 ± 0.017

Logistic Regression achieved the highest mean accuracy, followed closely by Random Forest. Both models clearly outperformed the Dummy Classifier. However, the more complex Random Forest model did not outperform the simpler baseline.

3. Conclusion

The results demonstrate why a complex machine learning model should always be compared with a simple baseline. Although Random Forest achieved strong predictive performance, Logistic Regression produced slightly better accuracy and offered greater simplicity and transparency. Therefore, model complexity alone does not guarantee improved performance.

For the doctoral research, Random Forest remains a relevant candidate for modeling complex relationships among academic, administrative, contextual, and temporal variables. However, it should only be preferred over simpler alternatives if it demonstrates consistent improvements in discrimination, calibration, fairness, performance under temporal validation, and practical usefulness. The final comparison must use institutional student data and a validation design that prevents data leakage and reflects the model's intended application.

TASK 2: PUBLISHED PAPER REPRODUCIBILITY AUDIT

1. Selected Paper

The selected study was Albugami, Wali, and Almagrabi (2026), *Predicting student dropout in Saudi Universities using machine learning and explainable AI*. The authors reported that KNN combined with Recursive Feature Elimination using 22 features achieved an accuracy and F1 score of 97.6%.

The paper was selected from the ten studies examined during the systematic literature review because it was closely aligned with the doctoral research topic and provided access to both data and code. The source code was downloaded as a ZIP file from the corresponding Zenodo repository, while the CSV dataset was obtained from the Supplemental Information section of the PeerJ article page.

These resources were used to reconstruct the analysis in the attached Task_2_Published_Paper_Reproducibility_Audit.ipynb notebook. The paper, dataset, and original code archive are included in the Task_2_Paper_Reproduction_Materials folder to document all materials used in the audit.

2. Reproduction Process

The dataset contained 4,560 observations and 26 variables. After removing one observation with a missing outcome and 37 duplicated records, 4,522 observations remained. The published procedure encoded the categorical variables, applied SMOTEENN, selected 22 features through RFE, divided the resulting data into 80% training and 20% testing sets, and trained a KNN classifier with five neighbors.

The repository did not provide the ordinal encoding definitions or the trainandtest() evaluation function. These components had to be reconstructed. In addition, SMOTEENN did not include a random seed, which prevented exact repetition of the sampling process.

3. Results

Evaluation	Accuracy	F1 score	ROC AUC
Published result	97.60%	97.60%	Not reported as the main claim
Reconstructed published workflow	95.63%	94.82%	98.44%
Corrected workflow	77.57%	55.77%	76.04%

In the corrected workflow, the original data were divided before applying SMOTEENN and RFE. Both procedures were fitted only to the training data. Accuracy consequently decreased to 77.57%, which was nearly equal to the 77.53% proportion represented by the majority class.

4. Three Line Report

Same number? No. The reconstructed workflow achieved 95.63% accuracy rather than the reported 97.6%, while the corrected evaluation achieved only 77.57%.

Valid? No. The reported performance is not a valid estimate for unseen students because SMOTEENN and RFE were applied before the training and test split, allowing test data to influence model development.

Problems? The repository omits essential definitions and functions, does not set a seed for SMOTEENN, performs resampling and feature selection before data splitting, uses arbitrary numeric encoding for nominal variables, and lacks repeated or external validation.

5. Conclusion

The study is partially reproducible because the published data and code allowed reconstruction of a result close to the reported accuracy. However, its main performance claim is not methodologically validated. The large reduction under the corrected evaluation indicates that the published 97.6% was probably inflated by data leakage and should not be interpreted as expected performance for new students.

6. References

Albugami, S., Wali, A., & Almagrabi, H. (2026). Predicting student dropout in Saudi Universities using machine learning and explainable AI. *PeerJ Computer Science, 12*, e3490. https://doi.org/10.7717/peerj-cs.3490

Albugami, S., Wali, A., & Almagrabi, H. (2026). *Saudi student dropout dataset* [Data set]. Supplemental material for *PeerJ Computer Science*. https://doi.org/10.7717/peerj-cs.3490/supp-3

shahadalbugami. (2025). *shahadalbugami/Predicting-Student-Dropout: Predicting student dropout with machine learning and explainable AI: A case study of Saudi universities* (Version V1) [Computer software]. Zenodo. https://doi.org/10.5281/zenodo.17069882